

AI Transparency by Design: A Mixed-Methods Evaluation of Trust and Adoption in AI-Powered Learning Management Systems

Aven Pacha Pinto

Southeast Missouri State University

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Dr. Mario Guimaraes

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Agenda

- ▶ Background and motivation
- ▶ Connection to the original study
- ▶ Problem statement and research objectives
- ▶ Literature review
- ▶ Proposed methodology
- ▶ Flowchart of the proposed method
- ▶ Expected results and discussion
- ▶ Conclusion and future work

Background and Motivation

AI in Learning Management Systems

- ▶ AI is increasingly used for:
 - ▶ Content generation
 - ▶ Personalized recommendations
 - ▶ Automated assessment and feedback
 - ▶ Intelligent tutoring
 - ▶ Adaptive learning
- ▶ Technical effectiveness does not guarantee user trust.
- ▶ Users may not know:
 - ▶ When AI was used
 - ▶ Why an output was produced
 - ▶ Which sources support it
 - ▶ How reliable the output is
- ▶ **Main concern:** AI capabilities may advance faster than users' ability to understand and evaluate them.

Connection to the Original Study

Original Study: Tian and Choo (2025) [2]

- ▶ Developed an educational content auto-generation system.
- ▶ Combined:
 - ▶ Large language models
 - ▶ Knowledge graphs
 - ▶ Learner modeling
 - ▶ Adaptive reasoning
- ▶ Reported improvements in:
 - ▶ Content quality
 - ▶ Knowledge coverage
 - ▶ Personalization
 - ▶ Generation efficiency
 - ▶ Student engagement

Extension
proposed in
this study

From: “Can generative AI improve educational content?”

To: “How should AI-powered LMS platforms communicate AI use so that users can evaluate and adopt it responsibly?”

Problem Statement and Research Gap

Problem Statement

- ▶ AI-powered LMS platforms may provide useful and technically advanced features while failing to clearly communicate how their AI processes work.

Research Gap

- ▶ Existing studies frequently focus on:
 - ▶ Technical performance
 - ▶ Personalization
 - ▶ Learning outcomes
 - ▶ General AI acceptance
 - ▶ Back-end explainability
- ▶ Fewer studies directly compare how visible AI-transparency UX/UI features affect:
 - ▶ User trust
 - ▶ Adoption intention
 - ▶ Responsible evaluation of AI-generated information

Objectives, Research Questions, and Hypotheses

Research Objectives

1

Evaluate the strength of AI-transparency UX/UI features in five AI-powered LMS platforms.

2

Examine the relationship between transparency, trust, and adoption intention.

3

Determine how general usability may influence adoption.

4

Explore how students and instructors interpret individual transparency features.

RQ1: Are stronger AI-transparency UX/UI features associated with higher trust and adoption-intention ratings?

RQ2: How do students and instructors explain the effects of AI labels, explanations, source visibility, and uncertainty indicators?

H₀: Transparency strength is not significantly associated with trust or adoption intention.

H₁: Stronger transparency is significantly associated with higher trust and adoption intention.

Literature Review Summary, Part I

Study	Focus and Method	Main Finding	Gap or Relevance
Luo et al. (2025) [1]	Systematic review of 63 higher-education AI studies	AI supports feedback, assessment, tutoring, and personalization	Calls for transparent, human-centered AI design
Tian and Choo (2025) [2]	Experimental evaluation of generative-AI educational content	Improved content quality, personalization, and efficiency	Did not measure user-facing transparency or trust
Ezzaim et al. (2024) [9]	Moodle adaptive-learning plugin; 102 students	Improved engagement, performance, and willingness to reuse	Transparency features were not isolated
Yang and Luo (2025) [8]	Design of an AI-based course teaching system	Demonstrated AI integration into course workflows	Limited evidence about trust or transparency
Kim et al. (2024) [6]	Taxonomy of generative-AI UI/UX patterns	Interfaces shape user understanding, control, and evaluation	Did not test university LMS trust or adoption

Literature Review Summary, Part II

Study	Focus and Method	Main Finding	Gap or Relevance
Villegas-Ch et al. (2025) [7]	Explainable Moodle assistant; 90 students	Demonstrated traceable AI feedback using LIME and SHAP	Focused mainly on technical performance
Marmolejo-Ramos et al. (2026) [3]	1,906 students across 13 countries	Trust, familiarity, effectiveness, and anxiety influenced adoption	Examined general AI rather than LMS interface features
Dang and Li (online 2025) [4]	Systematic review of 562 trust-in-AI studies	Transparency and explainability are important trust factors	Broad review, not specific to educational LMS interfaces
Colbran et al. (2026) [5]	Mixed-methods study of 121 students	Students valued AI but reported UX, ethics, and policy concerns	Studied chatbots rather than complete LMS platforms
Gadzhimusieva et al. (2026) [10]	Pilot evaluation with 30 teachers	Usefulness and ease of use influenced behavioral intention	Specialized context; transparency was not the main variable

Literature Synthesis and Remaining Gap

Four Main Conclusions

1. AI can improve personalization, feedback, content generation, and learning support.
2. Trust and adoption depend on human factors, not only technical performance.
3. Poor UX and uncertainty can reduce confidence in useful AI tools.
4. Transparency can be incorporated into AI interfaces and LMS workflows.

Remaining Gap

It is still unclear whether AI-powered LMS platforms with stronger combinations of user-visible transparency features receive higher trust and adoption ratings.

Proposed Contribution

- ▶ Compare five LMS platforms
- ▶ Apply one standardized transparency rubric
- ▶ Measure trust, adoption, and usability separately
- ▶ Use interviews to explain quantitative patterns

Proposed Research Design

Explanatory Sequential Mixed-Methods Design

QUAN → qual [11], [12]

Quantitative Phase

- ▶ Repeated-measures, within-subject comparison
- ▶ Every participant evaluates all five platforms
- ▶ Measures transparency, trust, adoption, and usability

Qualitative Phase

- ▶ Semi-structured follow-up interviews
- ▶ Selected participants explain high, low, or conflicting ratings

Why Mixed Methods?

- ▶ Quantitative data identify patterns.
- ▶ Qualitative data explain why those patterns occurred [11], [12].

Participants, Platforms, and Tasks

Participants

- ▶ Approximately 10-15 university students
- ▶ Approximately 3-5 university instructors
- ▶ At least 18 years old
- ▶ Prior experience using an LMS
- ▶ Convenience sampling

Coursebox AI

LearnWorlds

Mindsmith

TalentLMS

Moodle with
selected AI
plugins

Standardization

- ▶ Same sample course
- ▶ Same learning objectives
- ▶ Similar AI prompts and activities
- ▶ Fictional users and researcher-created data
- ▶ Counterbalanced platform order

Variables and Data Collection Tools

Independent Variable

AI-transparency UX/UI strength:

1. AI labeling
2. Explanations
3. Source visibility
4. Uncertainty indicators

Each component scored from 1 to 5.

Dependent Variables

- ▶ User trust
- ▶ Adoption intention

Additional Measure

- ▶ System Usability Scale, or SUS [13]
- ▶ Ten items converted to a 0-100 score

Other Data

- ▶ Background questionnaire
- ▶ Observation form
- ▶ Participant comments
- ▶ Semi-structured interviews

Procedure and Data Analysis

Procedure

1. Obtain informed consent.
2. Administer background questionnaire.
3. Introduce standardized tasks.
4. Test all five platforms in counterbalanced order.
5. Collect trust, adoption, and SUS ratings after each platform.
6. Select six to eight participants for interviews.
7. Integrate quantitative and qualitative findings.

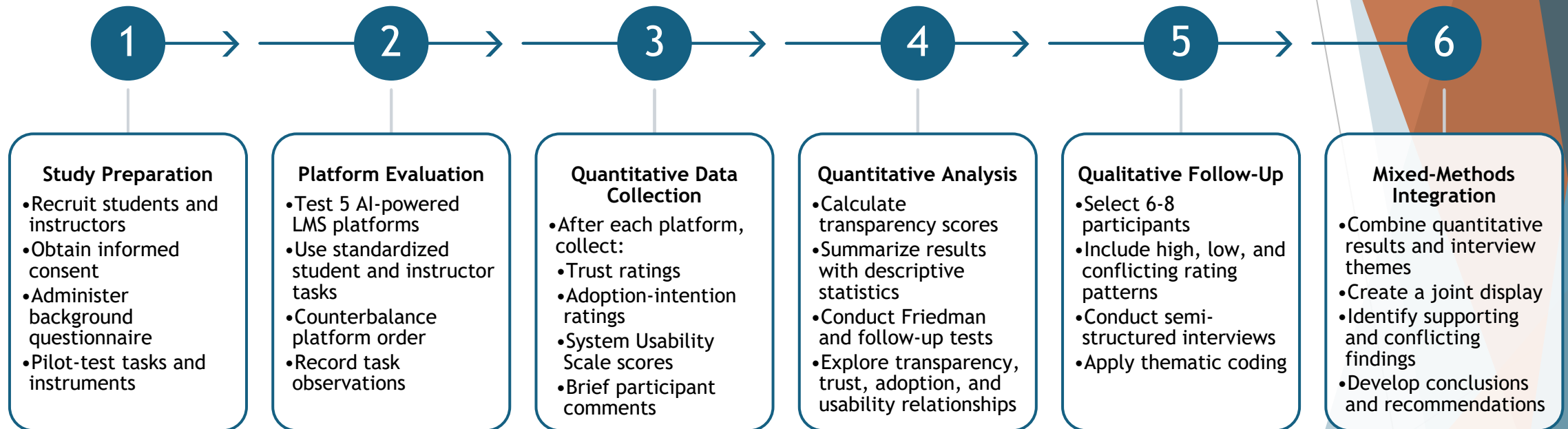
Quantitative Analysis

- ▶ Descriptive statistics
- ▶ Friedman tests
- ▶ Wilcoxon signed-rank follow-up tests
- ▶ Exploratory linear mixed-effects models
- ▶ Significance level: $\alpha = .05$

Qualitative Analysis

- ▶ Deductive and inductive thematic coding
- ▶ Comparison of high, low, and conflicting cases

Proposed Method



Expected Results & Discussion

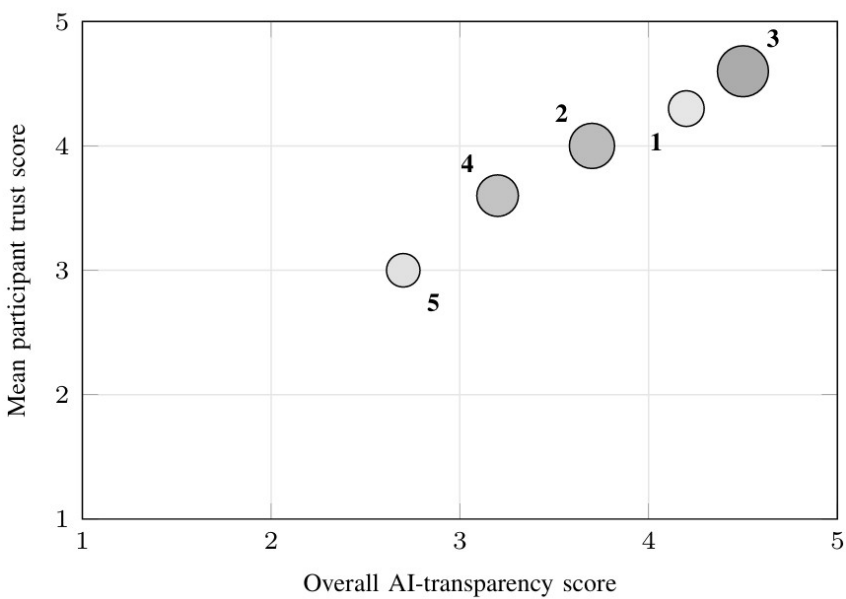
- Stronger AI transparency → higher trust
- Adoption influenced by both transparency and usability

Practical Contribution

- Treats AI transparency as a measurable UX/UI characteristic
- Supports LMS design, evaluation, and procurement decisions

Limitations and Ethical Considerations

- Small convenience sample and platform differences
- Findings show association, not causation
- IRB review, informed consent, fictional data, and privacy protections



Visual encoding: Larger bubbles indicate higher mean adoption intention. Darker bubbles indicate higher mean SUS scores.

#	Platform	Adoption	SUS	Illustrative QUAL Explanation
1	Coursebox AI	3.1	54	Clear AI features supported trust, but difficult navigation reduced willingness to adopt the platform.
2	LearnWorlds	4.1	76	Visible sources increased credibility and helped participants verify AI-generated information.
3	Mindsmith	4.5	82	Clear explanations made AI output easier to understand and evaluate.
4	TalentLMS	3.7	73	Participants found the interface usable but wanted more explanation of how AI results were produced.
5	Moodle with AI	2.9	58	Participants valued the system's flexibility but found its configuration confusing.

Illustrative joint display; all results are hypothetical. Integration approach based on [11], [12]; SUS measured using [13].

Conclusion

- ▶ AI transparency should be treated as a visible and measurable design characteristic.
- ▶ Stronger transparency is expected to support trust and adoption.
- ▶ Transparency does not operate independently of usability.
- ▶ Mixed methods can reveal both rating patterns and the reasons behind them.
- ▶ Responsible AI design should promote calibrated trust rather than blind acceptance.

Future Work



Test individual transparency features within the same LMS.



Conduct a larger multi-university study.



Follow students and instructors over an academic term.



Measure actual use, verification behavior, continued adoption, and changes in trust.

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